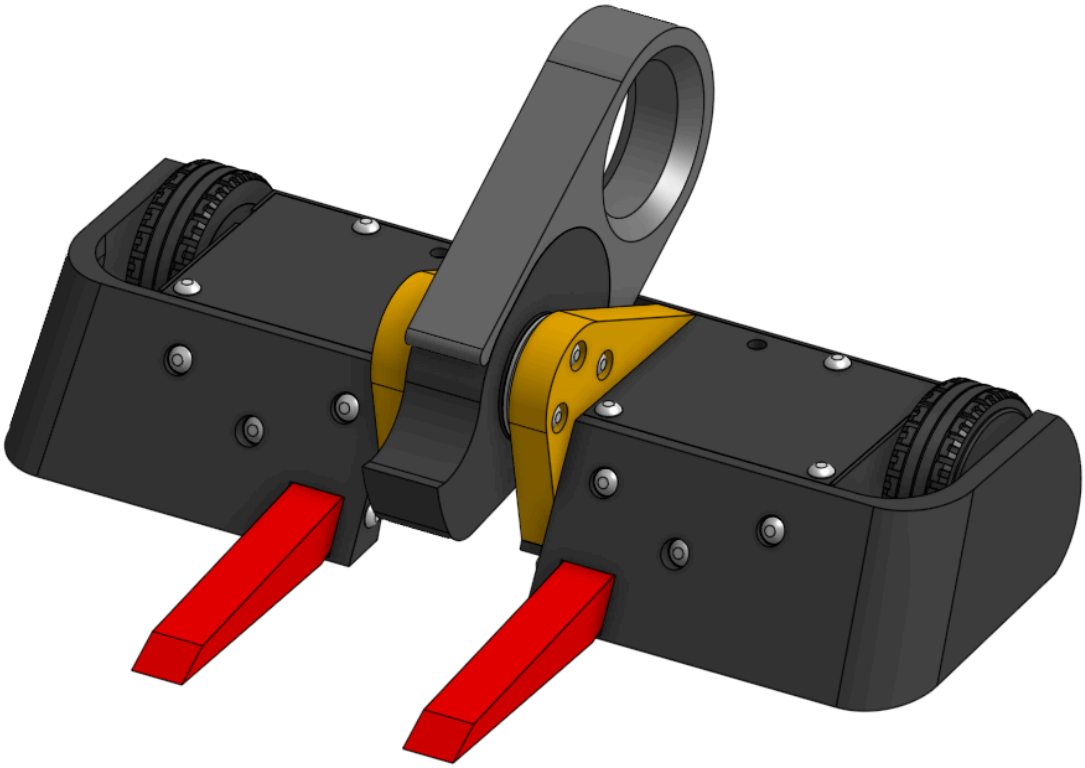
SLIDER V1 ASSEMBLY GUIDE



Designed by Mason Lu

REQUIRED SUPPLIES

All parts can be obtained through a single Repeat Open Plant Bundle.

Note: Slider was designed with the 3S 380 mAh GNB battery and F-06A Receiver. Other batteries and receivers may be used, but are not guaranteed to fit.

Electronics:

- 2x Mini Brushed Mk2 4mm shaft motors
- 1x Budget Ant Dual ESC
- 1x 2822 V3 2300kV
- 1x Blheli_s 35A Spinner esc
- 1x Repeat Screw Switch
- 1x 3S 380mAh GNB LiPo
- 1x F-06A Receiver

Hardware:

- 2x Gold Screw Hubs
- 2x Lego 43.2x14 Tire
- 1x HK0408 Needle Bearing
- 10x M3 x 8
- 18x M3 x 8 Thread Cutting Screws
- 2x M3 x 25
- 2x M2 x 5

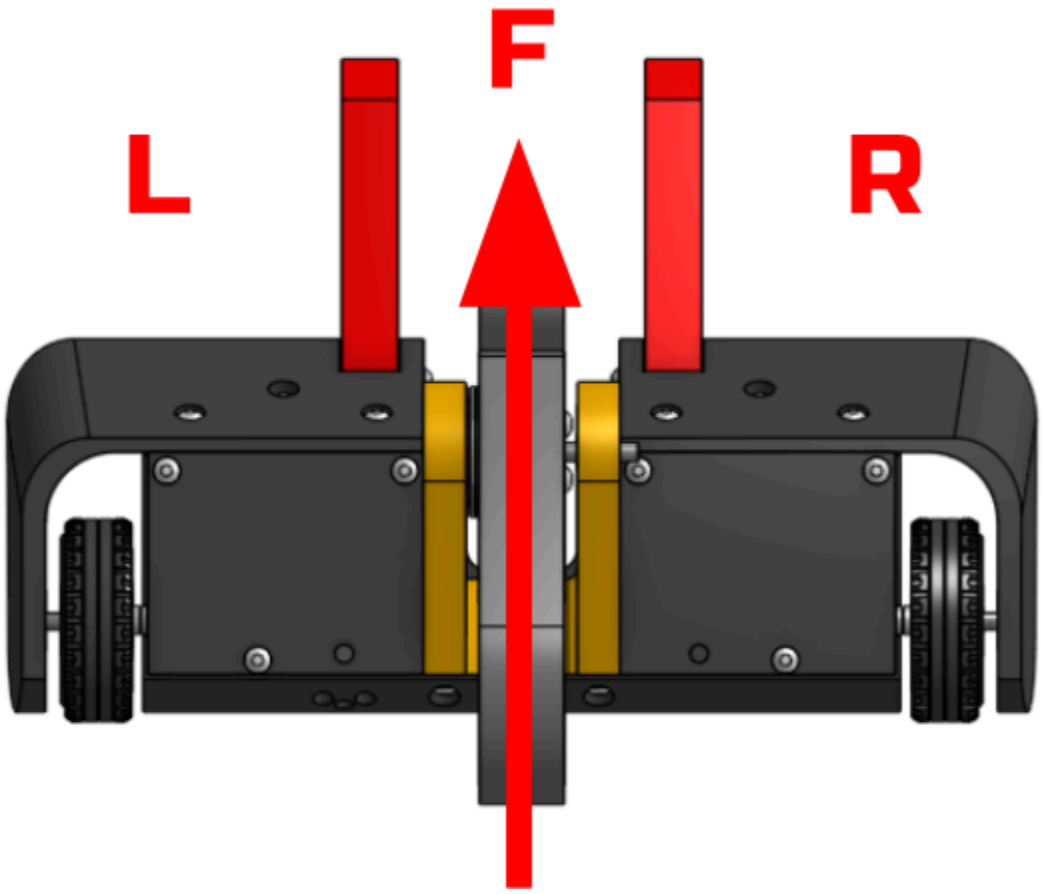
Printed Parts:

Note: It is recommended to print all parts in a variant of PLA+. Recommended slicer settings can be found in the weight allocation on page 5.

- 1x Chassis
- 2x Drive Motor Retainer
- 2x Lid
- 1x Motor Mount
- 1x Motor Support
- 1x Spinner
- 2x Wheel Hub
- 2x Fork
- 1x Armor Left
- 1x Armor Right
- 1x Weapon Lock

OVERVIEW

In this guide, left and right will be determined from a top-down view with the front of the robot pointing forward.



Soldering all electronics will take about 1 hour.

Assembling the robot itself should take around 10-15 minutes.

WEIGHT ALLOCATION

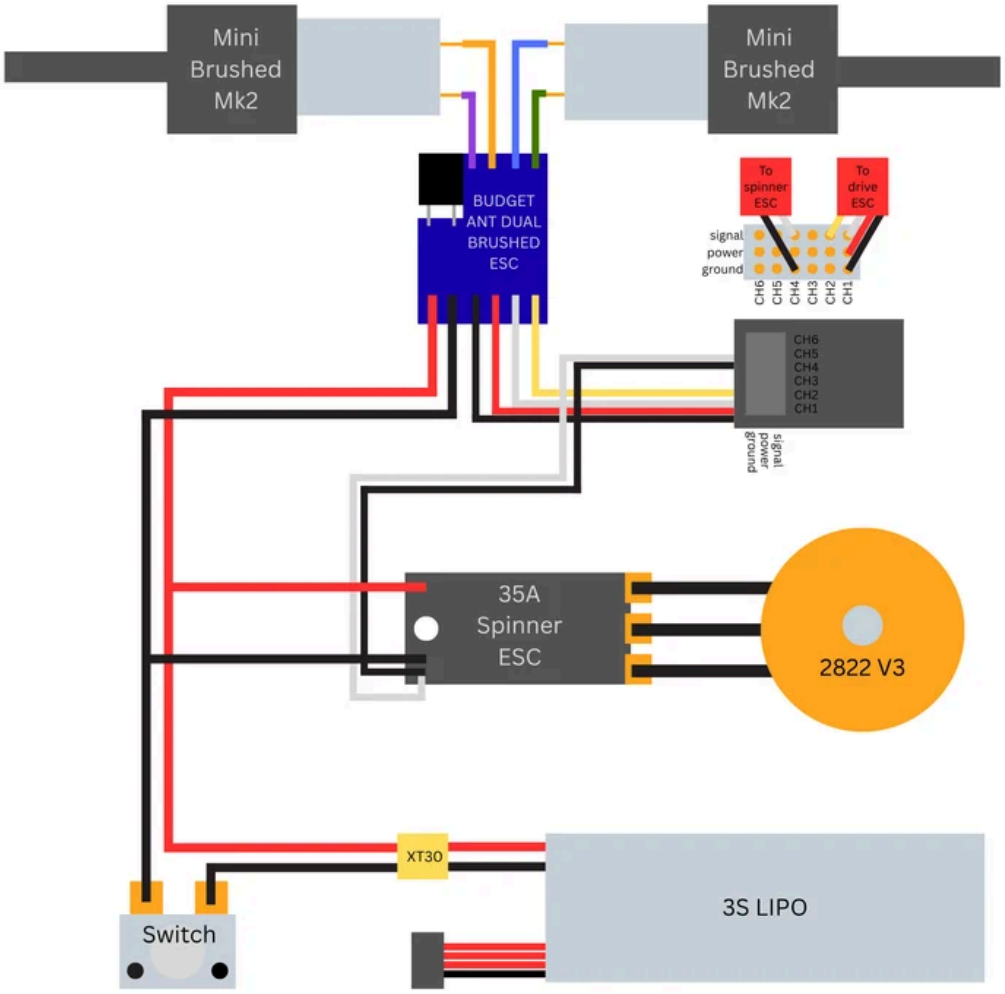
Total (g)	450.4								
Target (g)	453.592								
Remaining (g)	3.192								
Category	**Part**	**Weight Ea (g)**	**Qty**	**Subtotal**					
Electronics	Mini Brushed Mk2	27.5	2	55					
	Budget Ant Dual Drive Brushed ESC	7	1	7					
	Repeat 2822 Mk3	46	1	46					
	Blheli_s 35A Ant Spinner ESC	6	1	6					
	Repeat Screw Switch	1.5	1	1.5					
	F-06A Receiver	7.5	1	7.5					
	GNB 380mAh 3S LiPo	32	1	32					
	Wires	7	1	7					
Hardware	M3x8	0.65	10	6.5					
	M3x8 Thread Forming Screw	0.65	18	11.7					
	M2x5	0.2	2	0.4					
	HK0408 Needle Bearing	2.4	1	2.4					
	Gold Screw Hub	3.5	2	7					
	43.2x14 Lego Tire	8.6	2	17.2					

					Recommended print settings (0.4 mm nozzle)				
					Walls	**Top Shell**	**Bottom Shell**	**Infill (Gyroid)**	**Supports**
Printed Parts	Chassis	77	1	77	3	1 mm	1 mm	15%	No
	Spinner	45	1	45	2	N/A	N/A	100%	No
	MotorMount	15	1	15	4	2 mm	2 mm	20%	Yes
	MotorSupport	17	1	17	4	2 mm	2 mm	20%	No
	DriveMotorRetainer	2	2	4	3	1 mm	0.6 mm	15%	No
	43.2x14 Tire Hub	6	2	12	2	1 mm	0.6 mm	15%	Yes
	Lid	7	2	14	2	N/A	N/A	100%	No
Default Armor	ArmorLeft	24	1	24	3	1 mm	0.6 mm	15%	No
	ArmorRight	24	1	24	3	1 mm	0.6 mm	15%	No
	Fork	4	2	8	2	1 mm	0.6 mm	15%	No
	M3x25	1.6	2	3.2					
	(Custom armor should be under 60 g total)								

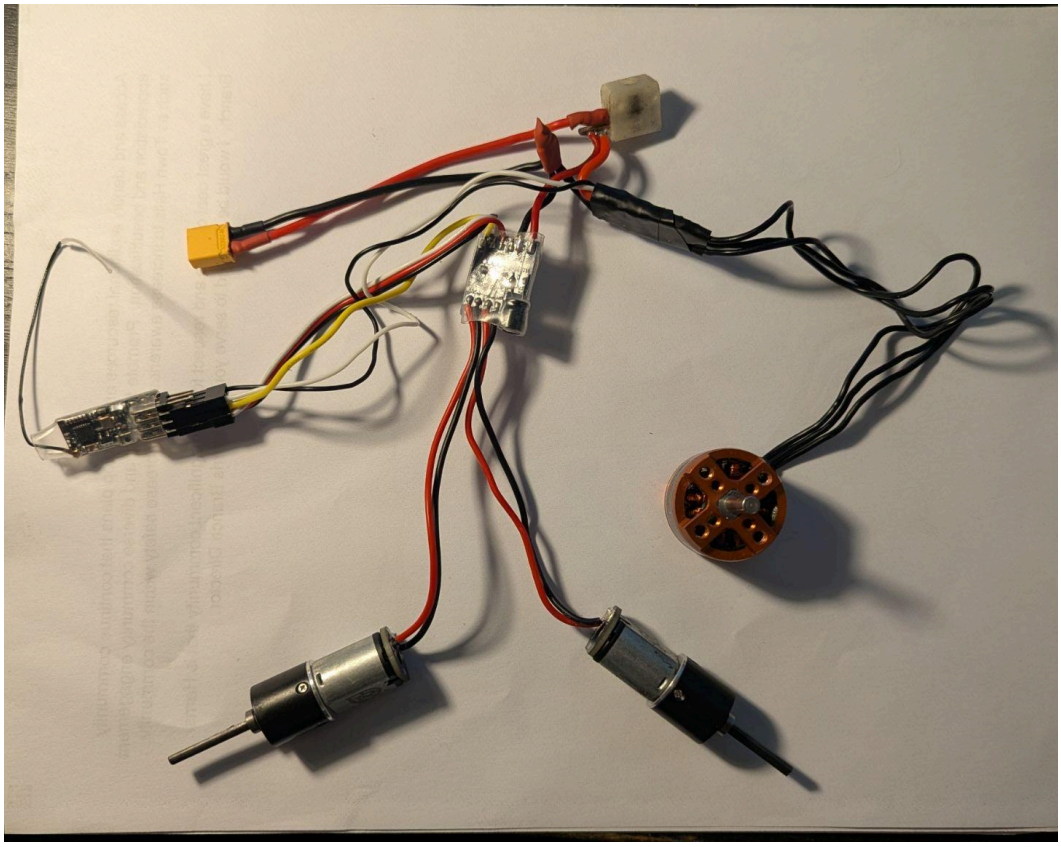
PRINTED PARTS

All printed parts are recommended to be printed in a variant of PLA+. Duramic PLA+ offers a good price-to-performance. If you want a tougher robot, Overture Super PLA+ can be slightly more shatter-resistant. I personally have been running all Duramic PLA+ since 2024 and have been really happy with how it performs.

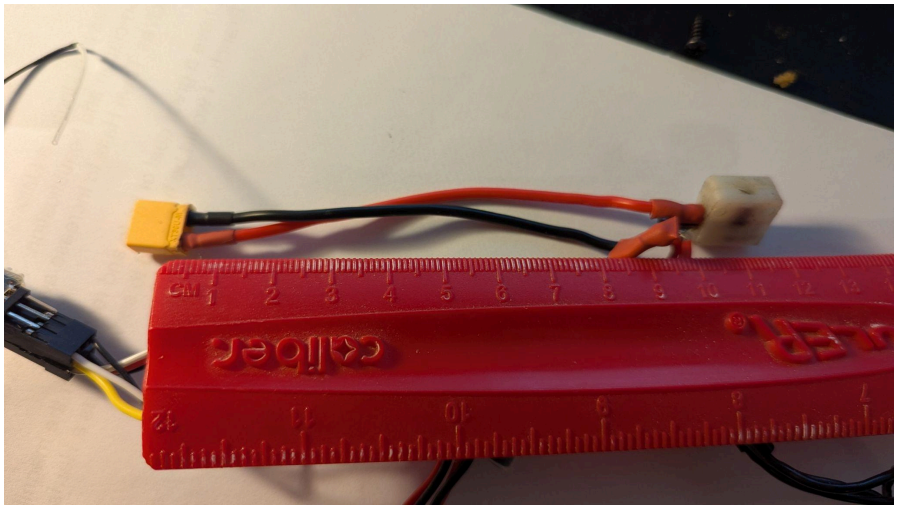
ELECTRONICS



- Solder/plug in the electronics in the Repeat Open Plant Kit as shown.
    - The wire between the switch and 3S LiPo should be at least 10 cm (3.5 in)
- Double-check your wiring and cover exposed components with heat shrink or electrical tape to ensure no shorts.
- Now is a good time to check which direction the orange weapon motor spins.
    - With the shaft facing up, it should spin counterclockwise.
    - If it is spinning clockwise, switch any two of the wires connecting the motor to the ESC.



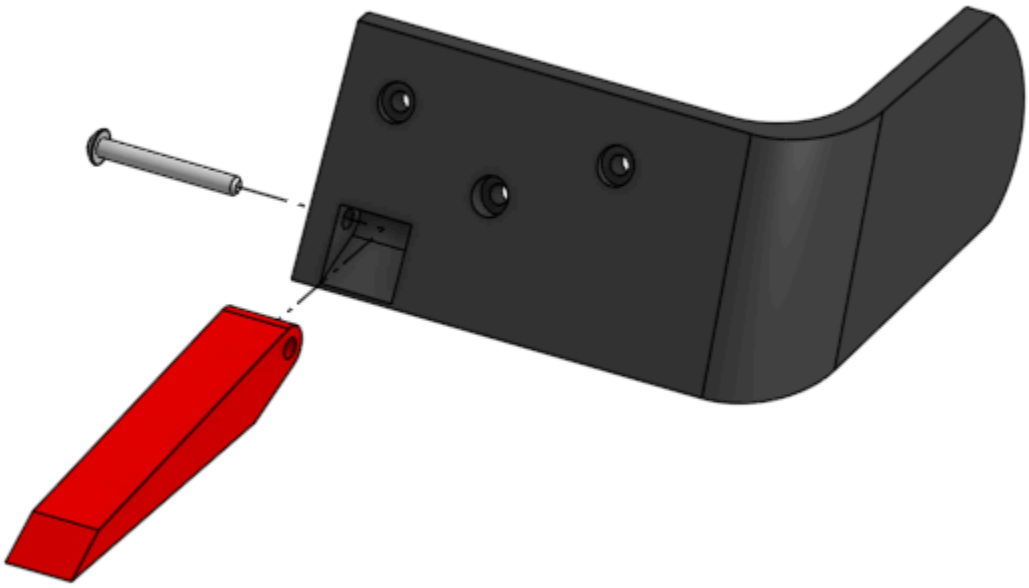
Example of soldered and connected electronics. Note that the receiver and switch are not the ones included in the Open Plant Kit (these are just what I had on hand). The



The wire from the 3S LiPo to the switch should be around 10 cm (3.5 in).

ASSEMBLY

ARMOR

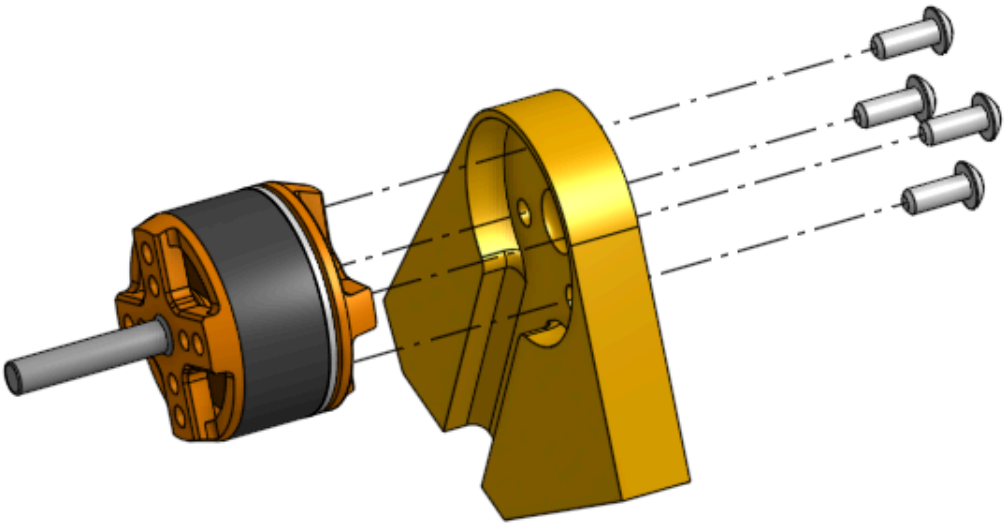


Insert a Fork into the Armor, and use an M3 x 25 to secure it in place. The fork should hinge on the screw and be slightly wiggly, but will not fall out.

Repeat for the other side.

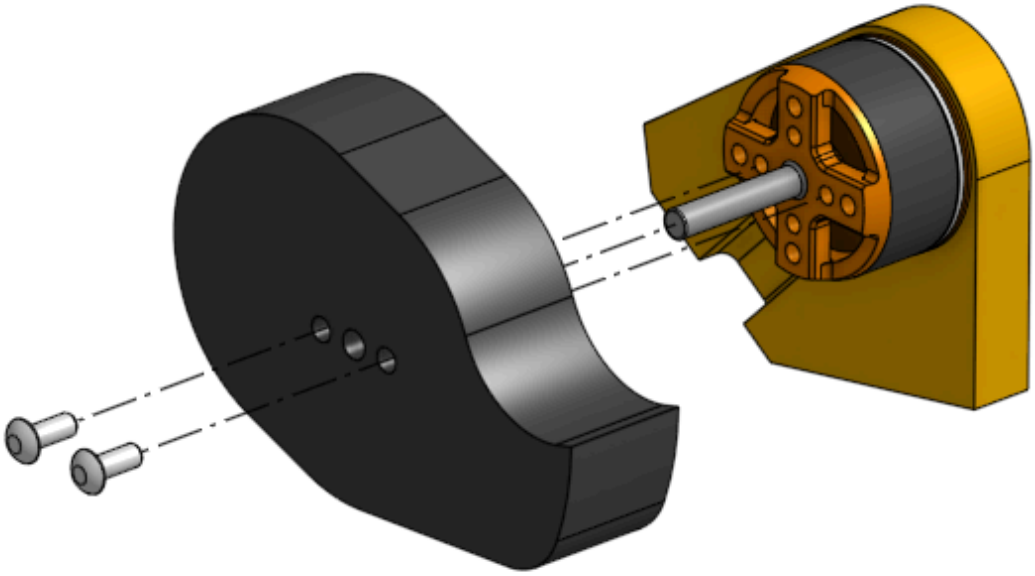
WEAPON STACK

STEP 1



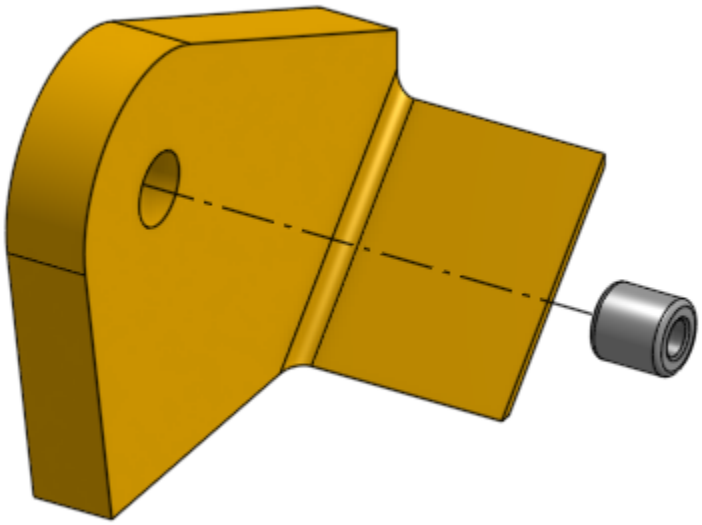
Secure the 2822 v3 into the Motor Mount with 4x M3 x 8 as shown. Ensure the motor is oriented such that the wires go through the wire channel.

STEP 2



Secure the Spinner to the 2822 v3 using 2x M3 x 8 screws. Set this assembly aside for now.

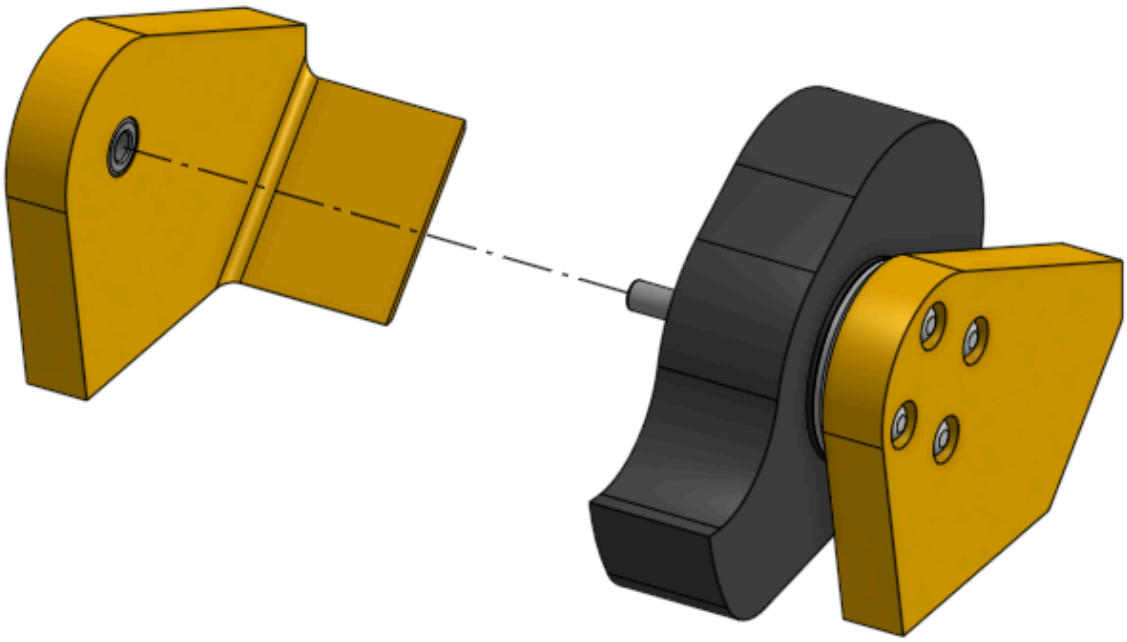
STEP 3



Insert the HK0408 Needle Bearing into the Motor Support. It will be a tight fit and may require a large amount of force.

Depending on your 3D printer's tolerances, it may be possible that the bearing will not fit. If the bearing still does not go in, it may be required to slightly sand down the inside of the bore. Similarly, if the bearing is too loose, add some superglue to hold it place, being careful not to get any on the inside of the bearing.

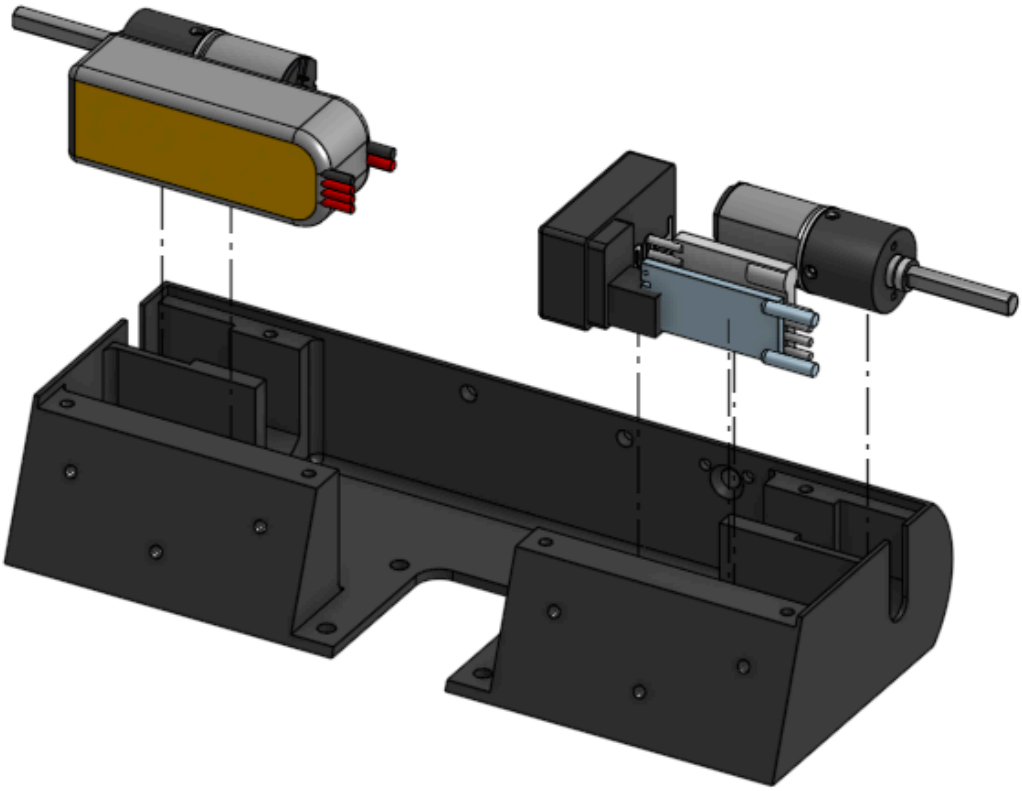
STEP 4



Insert the shaft of the 2822 v3 into the Needle Bearing. Set this assembly aside for now.

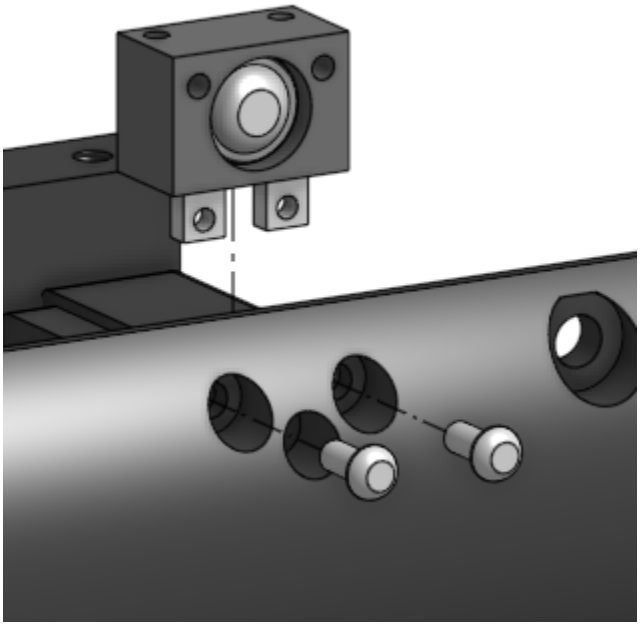
MAIN ROBOT

STEP 1



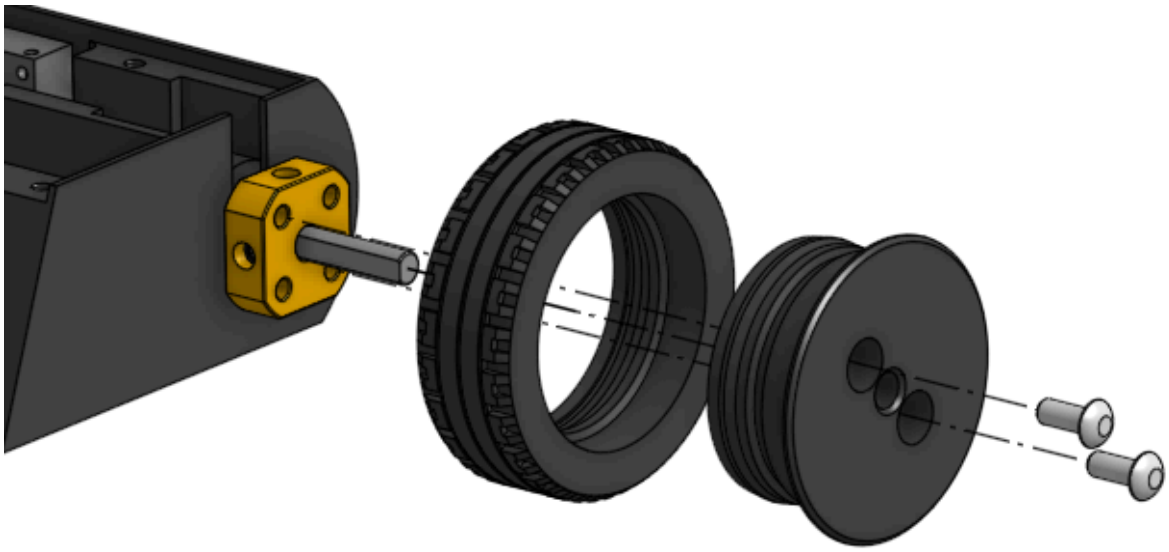
Place all electronics in the Chassis roughly as shown, with the battery on the right side and the remaining electronics on the left side. The Mini Brushed Mk2s should be a snug fit into the slots.

STEP 2



Install the Repeat Screw Switch as shown with two M2 x 5 screws.

STEP 3

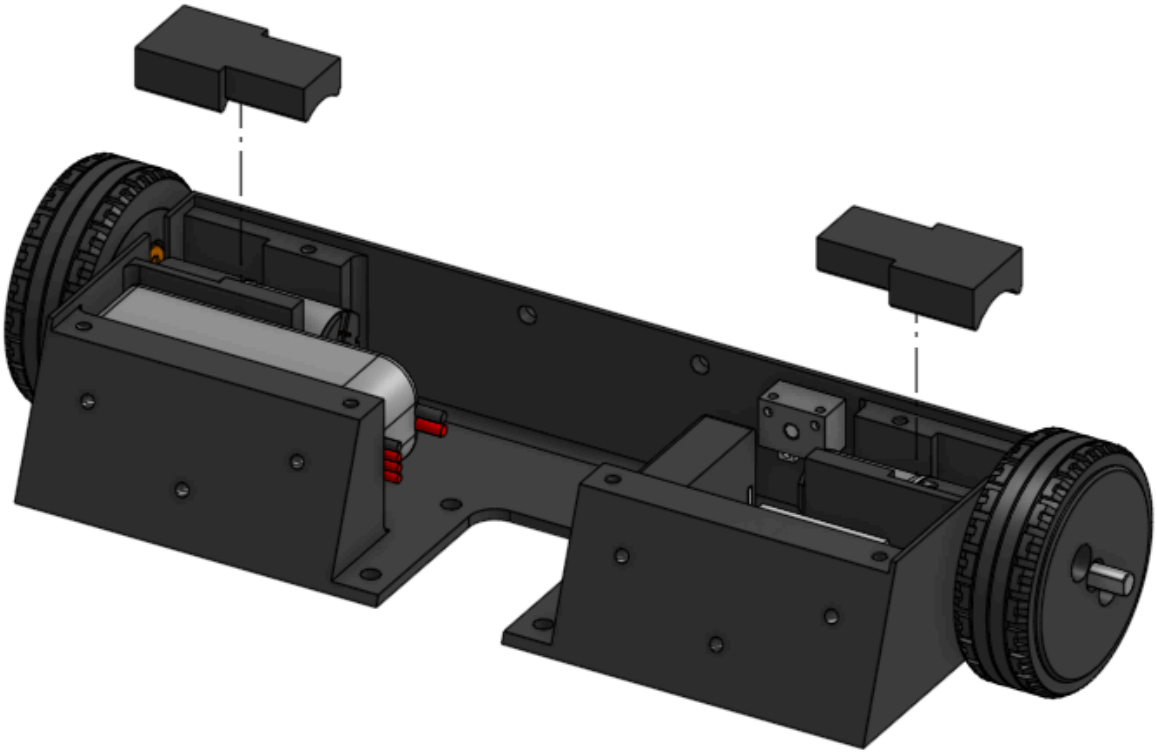


Slide on the Gold Screw Hub FIRST onto the Mini Brushed Mk2 and tighten down the set screws.

Slip the Lego 43.2 x 14 Tire onto the Wheel Hub as shown, and attach it to the Gold Screw Hub using 2x M3 x 8 screws.

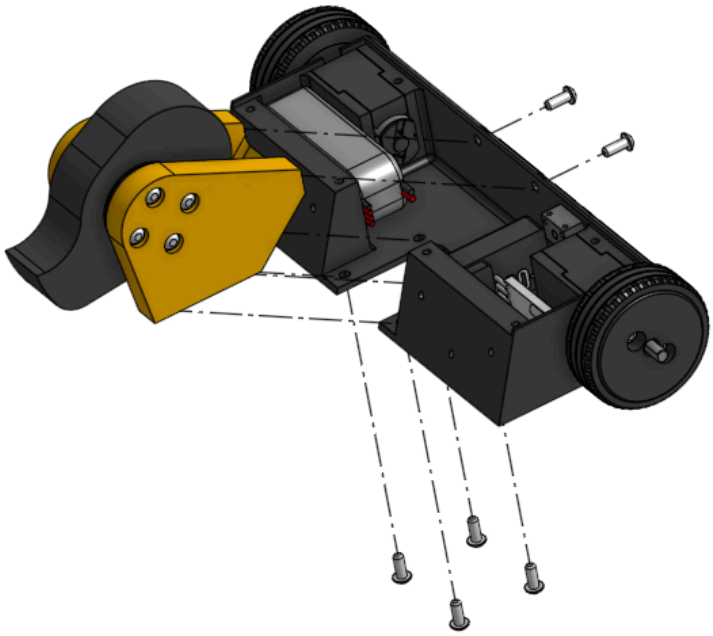
Repeat on the other side.

STEP 4



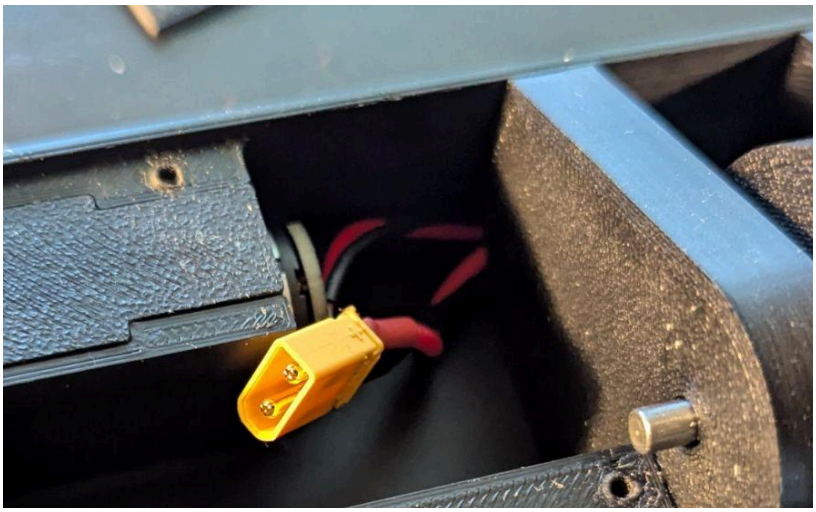
Insert the Drive Motor Retainers as shown.

STEP 5



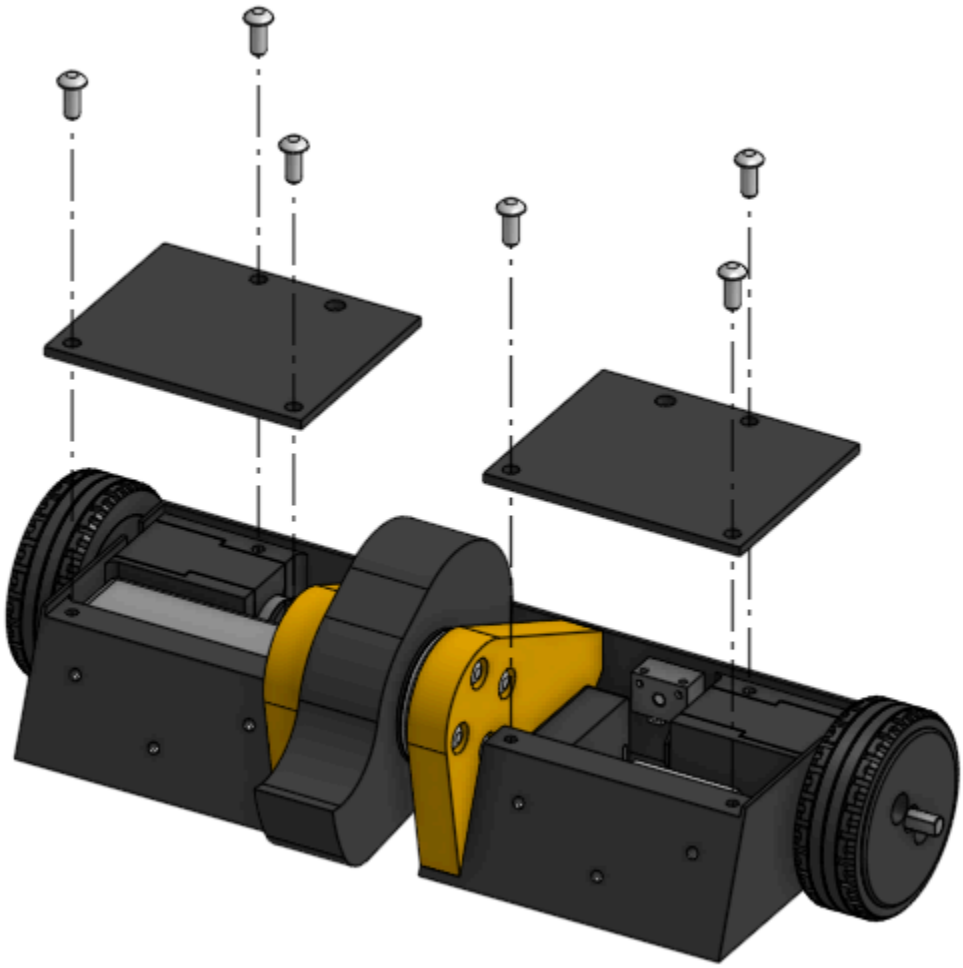
This is the most difficult step in the assembly process.

Insert the Weapon Stack assembly and insert it into the chassis, being careful not to pinch any of the wires. The battery connector wires and one pair of the drive motor wires should go along the back bottom corner of the chassis. Secure the assembly using 6x M3 x 8 Thread-Cutting Screws.



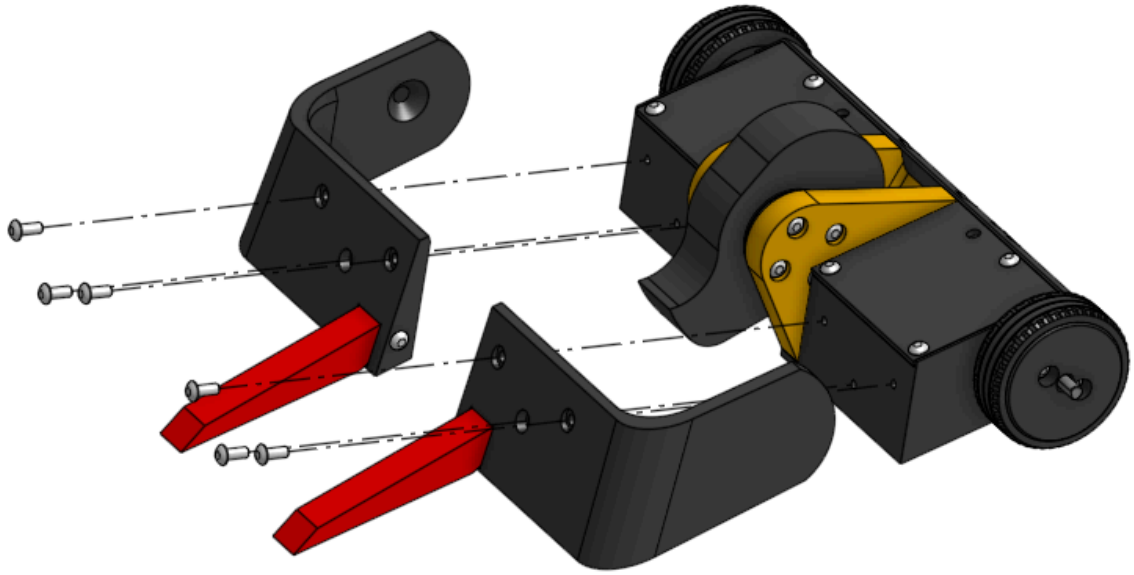
Example of wiring going to the right side of the robot.

STEP 6



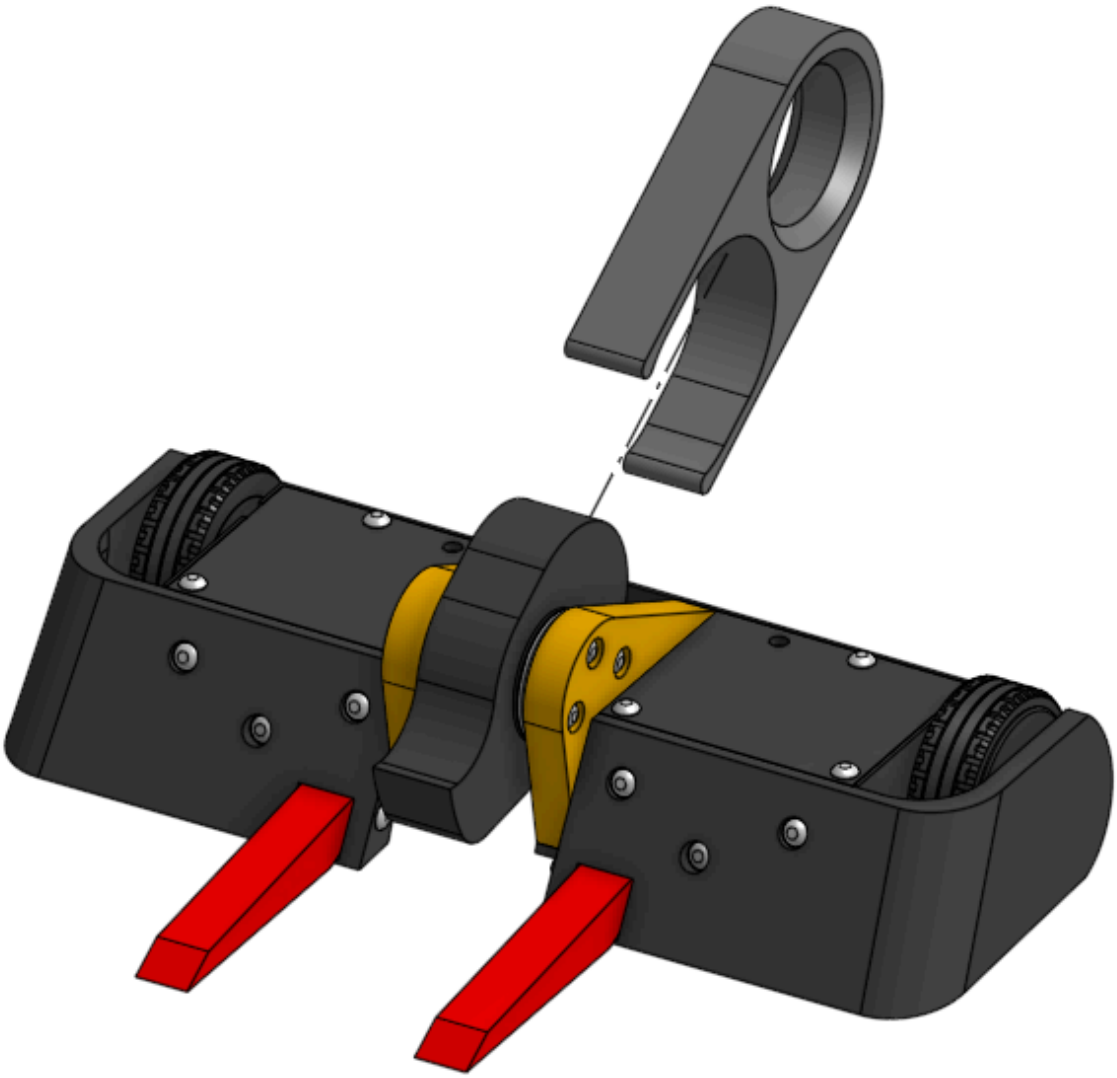
Arrange the electronics so that they comfortably fit in the left compartment. Attach 2x Lids using 6x M3 x 8 Thread-Cutting Screws.

STEP 7



Attach the armor packages as shown using 6x M3 x 8 Thread-Cutting Screws.

STEP 8



Slide on the Weapon Lock.

Congratulations! You have assembled a Slider! Test responsibly in a test box, and have fun competing with your new PLAntweight combat robot!

You can also customize the armor packages using the public .step files and Onshape assembly.